

Market Basket Analysis

961731 Business Data Analytics

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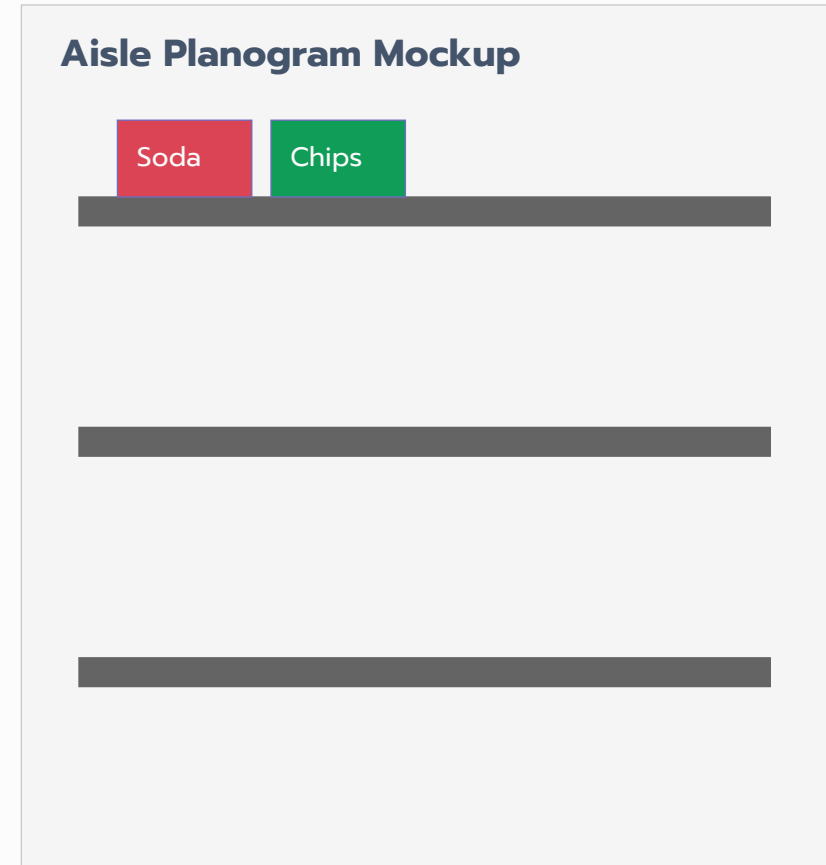
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Module 1: Strategic Motivation

Connecting cross-domain transactional frictions to enterprise capital efficiency

Systemic Retail Inefficiencies: Space & Flow

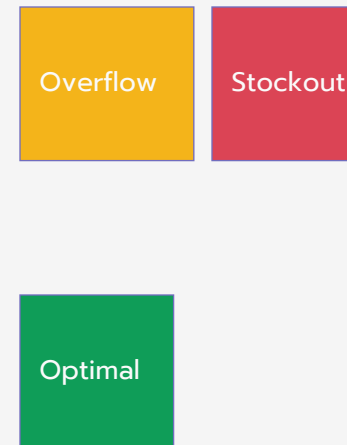
- Traditional retail layout:
 - Built on static vendor shelf-space negotiations rather than real customer baskets.
- Spatial bottlenecks:
 - Placing associated items apart increases shopper transit friction and checkout abandonments.



Systemic Retail Inefficiencies: Inventory Costs

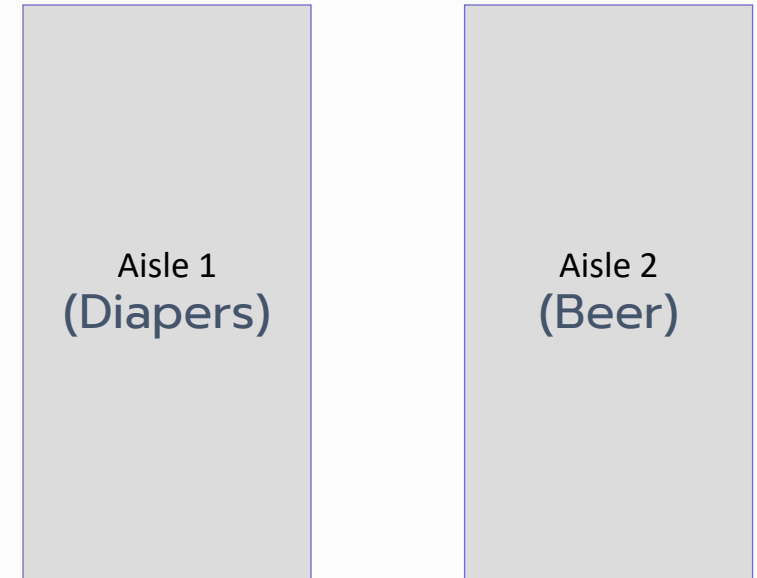
- Capital allocation lockup:
 - Over-stocking low-margin products that fail to generate pull effects for secondary categories.
- Operational risk profiles:
 - Stockouts of joint dependencies (e.g. burger buns without patties) decimate total ticket size.
- **JOINT PRODUCT OUT-OF-STOCKS**
 - Losing high-value baskets due to single accessory out-of-stocks.

Warehouse Storage Matrix



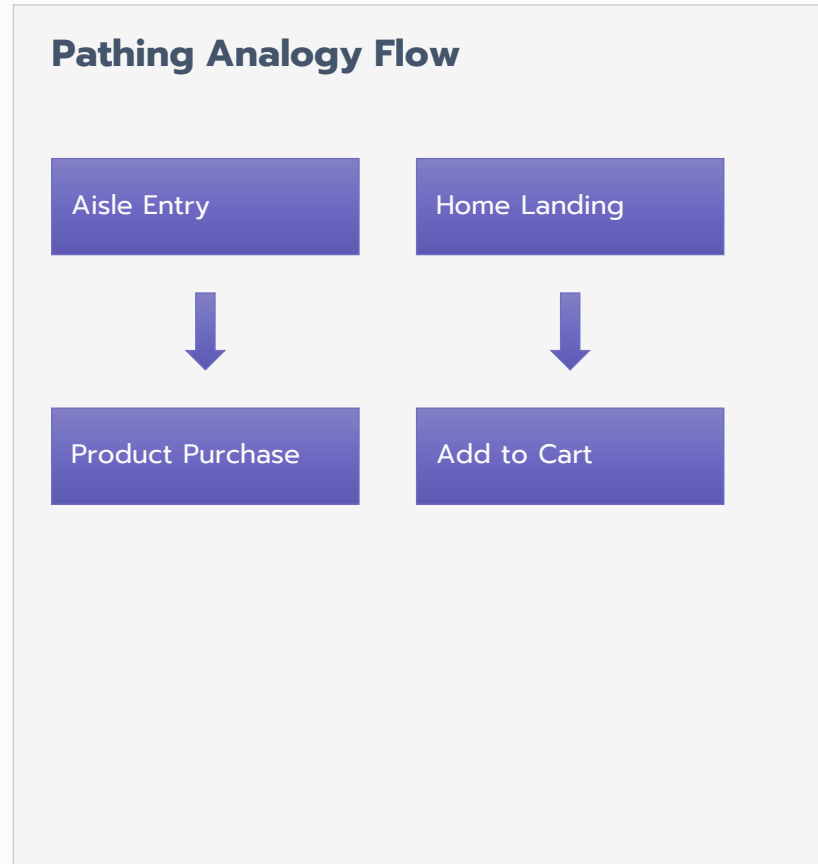
Convenience Store Maze

- Layout architecture
 - Store layouts are deliberately structured to maximize travel paths and co-occurrence exposure.
 - Failing to optimize shelf adjacent placement increases product search friction.
- Operational risk
 - Misplaced anchor items reduce average basket size below the target profit threshold.



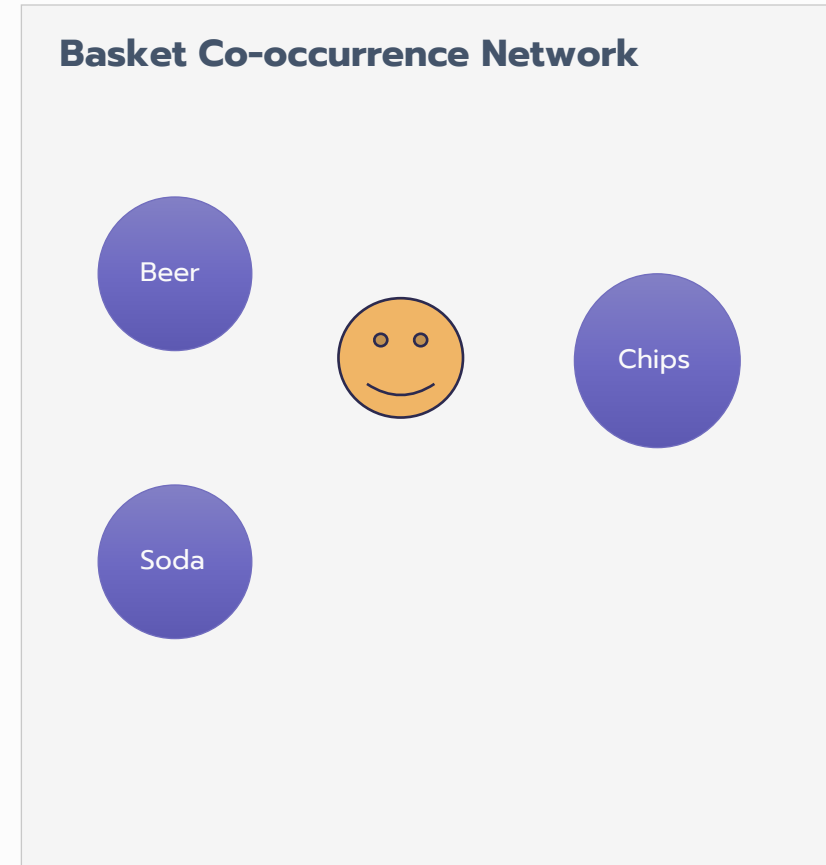
Digital Retail Scale

- Physical path coordinates:
 - Tracking visitor footprints through aisle layouts using smart cart sensors.
- Digital funnel mapping:
 - Applying clickstream logs to trace page navigation, add-to-carts, and checkouts.



The Co-occurrence Problem

- Item dependencies:
 - Consumer purchases are driven by systemic needs, not isolated decisions.
- Operational failure modes:
 - Standard single-item reports miss joint probability patterns across the catalog.



Module 2: Value Vectors

Quantifying affinity positioning, cross-selling, and upselling impact

Affinity Positioning

- Physical alignment
 - Placing high-affinity product groups adjacent to each other on shelves.
 - Reduces physical search times while increasing unplanned co-purchases.
- Strategic anchor
 - Using Beer and Potato Chips as classic high-support proximity placement anchors.

Cross-Merchandising Shelf

[Beer] <---> [Potato Chips]

Cross-Selling Mechanics

- Bundle pricing
 - Combining complementary products (e.g. burger and fries) into a single transaction SKU.
 - Increases transaction velocity and elevates the average margin per basket.
- Digital prompts
 - Providing automated checkout suggestions based on high-confidence association rules.

Shopping Basket

- Burger (\$5.99)

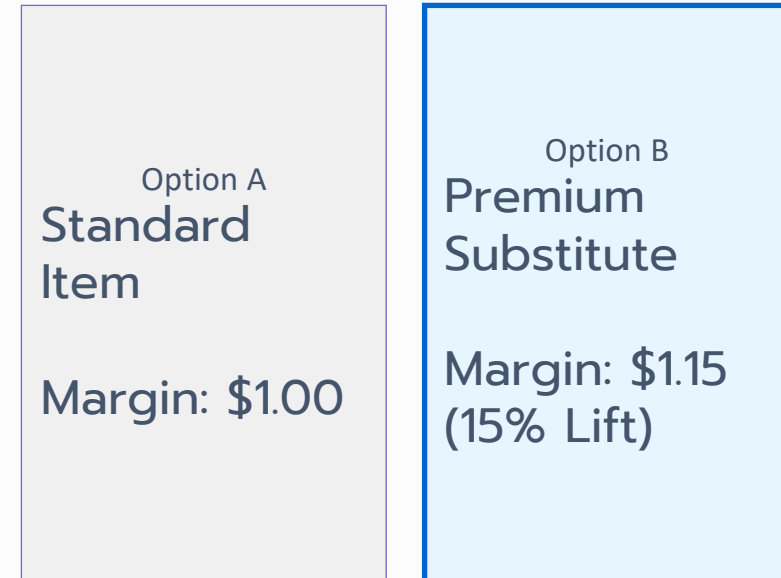
Recommended Add-On:

[+] French Fries (\$1.50)

Save 20% when bought together

Upsell Strategy

- High-margin substitutes
 - Directing customer purchase behavior towards a higher-priced alternative product.
 - Increases revenue capture and delivers a expansion in product category gross margin.
- Value optimization
 - Replacing commodity items with premium private-label substitutes in high-confidence baskets.



Value-Driver Tree

- Margin decomposition
 - Mapping store profit directly to basket size, volume, and affinity lift.
 - Provides a MECE framework to prioritize cross-merchandising over direct discount promotions.
- Strategic decision
 - Identify nodes where co-occurrence patterns capture high margins without cannibalizing baseline products.



Operational Metrics

- Supply chain staging
 - Positioning high-association items near each other in warehouses decreases order pick paths.
 - Improves order dispatch times and increases distribution center picker throughput.
- Stock management
 - Aligning safety stock levels of paired items prevents split-shipment delivery failures.

Warehouse Zone A
(Affinity Placed Items)

Reduces picking path
by 25%
Increases worker
throughput

Strategic Applications

- Seasonal adaptation
 - Updating store planograms dynamically based on seasonal affinity shifts.
 - Ensures shelf space is allocated to maximum probability transaction combinations.
- Inventory lifecycle
 - Synchronizing product discount timelines for paired items prevents margins from eroding prematurely.

Q1: Winter
**Warm
Beverages**

Q2: Spring
**Fresh
Produce**

Q3: Summer
**Beer &
Chips**

Q4: Autumn
**Holiday
Baking**

Revenue Optimization Value-Driver Tree

- EBITDA growth vector:
 - Revenue increase driven by larger Average Order Value (AOV) and customer retention.
- AOV optimization:
 - Leveraging cross-sell conversion rates to move secondary basket items.

Revenue Driver	Market Basket Impact
Average Order Value	Increase items-per-transaction
Inventory Cost	Optimize holding patterns
Customer Retention	Align store layout to shopping needs

Module 3: Traditional Methods

Exposing mathematical limitations and computational friction of cross-tabulation

Baseline Transaction Dataset

TxID	Items in Basket
1	Beer, Pretzels, Potato Chips, Aspirin
2	Diapers, Baby Lotion, Grapefruit Juice, Baby Food, Milk
3	Soda, Potato Chips, Milk
4	Soup, Beer, Milk, Ice Cream
5	Soda, Coffee, Milk, Bread
6	Beer, Potato Chips

Co-occurrence Matrix

- Cross-tabulation count:
 - Extracting joint frequencies by constructing a symmetric co-occurrence matrix.
- Symmetry condition:
 - Diagonals show single item frequencies; off-diagonals show joint counts.

Item	Beer	Chips	Milk	Diap	Soda
Beer	3	3	2	0	1
Chips	3	3	2	0	1
Milk	2	2	5	2	2
Diapers	0	0	2	2	1
Soda	1	1	2	1	2

Combinatorial Explosion

- Exponential growth
 - The number of unique itemset combinations grows exponentially at a rate of 2^d .
 - An item inventory of 1,000 SKUs generates 1.07×10^{301} possible combinations.
- Compute threshold
 - Processing large inventory transaction matrices causes severe **risk** of server memory crashes.

Combinatorial Growth

$d = 5 \Rightarrow 31$ sets

$d = 10 \Rightarrow 1,023$ sets

$d = 100 \Rightarrow 1.27 \times 10^{30}$ sets

Asymmetric Risk: server crash

Zero-Inflated Sparsity

- Matrix sparsity
 - Enterprise retail transaction databases exhibit matrix sparsity exceeding 99% zeros.
 - Standard relational databases waste significant memory holding null binary values.
- Strategic impact
 - Calculating co-occurrence across sparse matrices generates an **risk** of extreme computational overhead.

99.9% Zero-Inflated Matrix

```
[1, 0, 0, 0, 0, 0, 0]  
[0, 0, 0, 0, 1, 0, 0]  
[0, 0, 0, 0, 0, 0, 0]  
[0, 1, 0, 0, 0, 0, 0]
```

Correlation vs Association

Correlation vulnerabilities:

Standard correlation coefficients (like Pearson) are heavily biased by zero-inflated data.

Association logic:

Focuses exclusively on positive transaction events, ignoring global empty cells.

Methodological Comparison

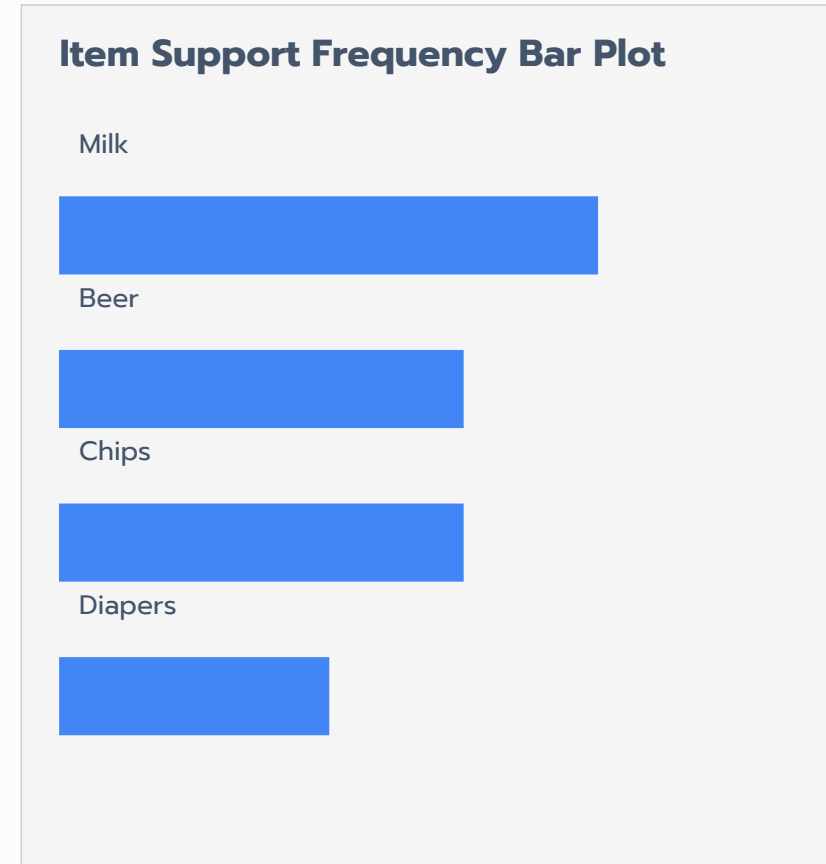
Metric	Traditional Cross-Tab	ML Apriori Association
Computational Complexity	$O(d^2)$ matrix sizing	$O(2^d)$ pruned tree sizing
Data Sparsity Sensitivity	High null-computation waste	Pruned support filtering
Actionable Output	Symmetric co-occurrence counts	Directional rules ($X \rightarrow Y$)

Module 4: Machine Learning Paradigm

Formalizing unsupervised association rule mining and core metrics

EDA: Item Frequencies & Long-Tail

- Pareto distribution:
 - The top 20% of products dominate more than 80% of transaction counts.
- Long-tail values:
 - Niche items have low support but can build highly profitable association rules.



Apriori Principle

- Downward closure
 - All subsets of a frequent itemset must also be frequent.
 - If an itemset is infrequent, all its supersets are immediately pruned.
- Pruning rules
 - Drastically reduces candidate space search size before counting transaction frequencies.

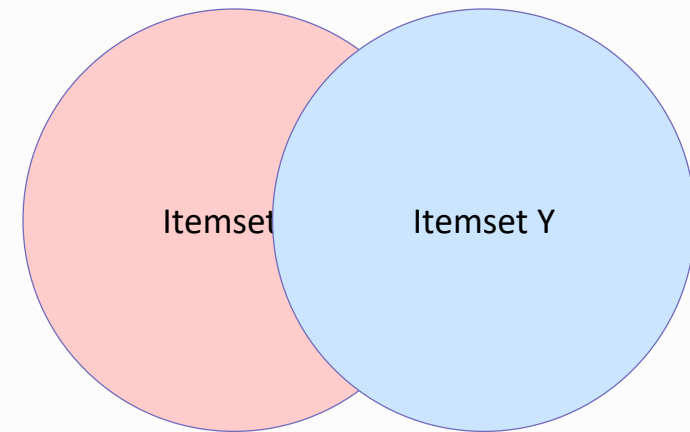
Apriori Principle

If $\{A, B\}$ is frequent,
then $\{A\}$ and $\{B\}$ are
frequent.

If $\{A\}$ is infrequent,
then $\{A, B\}$ is pruned.

Support Metric

- Frequency measure
 - Formula: $\text{Support}(X \rightarrow Y) = P(X \cup Y) = \text{count}(X \cup Y) / N$.
 - Represents the proportion of transactions containing both itemsets X and Y.
- Business meaning
 - Quantifies rule popularity. Sets threshold to discard low-velocity product combinations.



Confidence Metric

- Conditional probability
 - Formula: $\text{Confidence}(X \rightarrow Y) = P(Y \mid X) = \text{Support}(X \cup Y) / \text{Support}(X)$.
 - Measures the likelihood of purchasing item Y given item X is bought.
- Business meaning
 - Measures rule strength. Directs the targeted conversion rate for product recommendations.

Conditional Probability

$$P(Y \mid X) = P(X \text{ and } Y) / P(X)$$

Of those who buy X,
what % also buy Y?

Lift Metric

- Independence ratio
 - Formula: $\text{Lift}(X \rightarrow Y) = \frac{P(X \cup Y)}{(P(X) * P(Y))} = \frac{\text{Confidence}}{P(Y)}$.
 - Compares co-occurrence frequency to the probability of random independent purchases.
- Business meaning
 - Lift > 1 indicates positive association. Lift <= 1 implies random independence.

Lift Ratio

Lift > 1: Positive Association
 Lift = 1: Independent
 Lift < 1: Negative Association

Rule Interpretation Matrix

Lift Level	Support Level	Confidence Level	Strategic Action
Lift > 1.0	Support >= 5%	Confidence >= 30%	Anchor bundle at check-out
Lift > 1.0	Support < 5%	Confidence >= 40%	Niche targeted promotion
Lift <= 1.0	Any level	Any level	Discard rule (negative association)

Strategic Placement Playbook

- Product matrix
 - High support and high confidence rules are deployed as physical aisle anchors.
 - Low support and high confidence rules are targeted for digital cross-sell bundles.
- Risk mitigation
 - Deploying a promotion with lift < 1.0 creates a **stop-loss trigger** to immediately halt campaign.

Strategy 2x2 Matrix

- High Sup / High Conf => Anchor
- High Sup / Low Conf => Cross-sell
- Low Sup / High Conf => Bundle
- Low Sup / Low Conf => Discard

Module 5: Pedagogical Board Demo

Step-by-step transaction calculations and candidate itemset pruning

Step 1: Candidate 1-Itemsets

- Frequency counts
 - Calculate support for all single item candidates from transaction data.
 - Item count: Beer=9, Chips=10, Milk=12, Soda=7, Diapers=4.
- Initial matrix
 - Generates a base list of single-item occurrences across 20 transactions.

Step 1: Compute Support

{Beer}: $9/20 = 45\%$
 {Chips}: $10/20 = 50\%$
 {Milk}: $12/20 = 60\%$
 {Soda}: $7/20 = 35\%$
 {Diapers}: $4/20 = 20\%$

Step 2: Frequent 1-Itemsets

- Support pruning
 - Apply a minimum support threshold of 15% ($\text{min_count} \geq 3$).
 - All 5 single items exceed threshold and are retained in the frequent list.
- Itemset containment
 - Pruned frequent 1-itemset list L1 contains Beer, Chips, Milk, Soda, and Diapers.

Step 2: Pruning

Apply $\text{min_support} = 15\%$
All itemsets pass.
None are pruned.

Step 3: Candidate 2-Itemsets

- Join operation
 - Combine items in L1 to generate candidate pairs: {Beer, Chips}, {Beer, Milk}, etc.
 - Generates exactly 10 candidate pairs for support counting.
- Pair matching
 - Count occurrences of these pairs in the 20 transaction rows.

Step 3: Pair Candidates

Join frequent 1-itemsets:
{Beer, Chips}, {Beer, Milk},
{Chips, Milk}, {Soda, Milk},
{Diapers, Milk}

Step 4: Frequent 2-Itemsets

- Frequent pair pruning
 - Apply min_support = 15% to prune candidate pairs.
 - Pairs passing: {Beer, Chips}=6, {Soda, Milk}=6, {Diapers, Milk}=4.
- Frequent pairs list
 - Retained list L2 contains three strong co-occurrence transaction pairs.

Step 4: Frequent Pairs

{Beer, Chips}: $6/20 = 30\%$
{Soda, Milk}: $6/20 = 30\%$
{Diapers, Milk}: $4/20 = 20\%$

Step 5: Rule Extraction

- Confidence calculation
 - Extract rules from frequent pairs: {Beer} $\rightarrow$ {Chips} has confidence = $30\% / 45\% = 66.7\%$.
 - Lift is confidence / $P(\text{Chips}) = 66.7\% / 50\% = 1.33$.
- Association validation
 - All extracted rules are sorted by lift to identify top-performing cross-merchandising actions.

Rule Extraction

{Beer} $\rightarrow$ {Chips}

Support = 30%

Confidence = $30\% / 45\% = 66.7\%$

Lift = $66.7\% / 50\% = 1.33$

Extracted Association Rules

Rule (X -> Y)	Support(X U Y)	Confidence(Y X)	Lift(X -> Y)	Interpretation
{Beer} -> {Chips}	30.0%	66.7%	1.33	Positive association
{Soda} -> {Milk}	30.0%	85.7%	1.43	Positive association
{Diapers} -> {Milk}	20.0%	100.0%	1.67	Strong positive association

Module 6: Programmatic Mining

Python implementation of association mining using pandas and mlxtend

Preprocessing Code

```
import pandas as pd
from mlxtend.preprocessing import TransactionEncoder

df_raw = pd.read_csv('groceries.csv', header=None)
transactions = []
for i in range(len(df_raw)):
    row = [str(df_raw.values[i, j]) for j in range(df_raw.shape[1])]
    row_clean = [item for item in row if item != 'nan']
    transactions.append(row_clean)

te = TransactionEncoder()
te_ary = te.fit(transactions).transform(transactions)
df_encoded = pd.DataFrame(te_ary, columns=te.columns_)
```

- Code explanation
 - Lines 1-2: Import pandas and TransactionEncoder from mlxtend.
 - Line 4: Ingest transactional database with header=None.
 - Lines 5-9: Cleanse dataset, stripping null string values from transactions.
 - Lines 11-13: One-hot encode the transactions list into a binary matrix.

Apriori & Rules Code

```
from mlxtend.frequent_patterns import apriori, association_rules

# Run Apriori algorithm
frequent_itemsets = apriori(df_encoded, min_support=0.01,
                             use_colnames=True)

# Generate association rules
rules = association_rules(frequent_itemsets, metric='confidence',
                           min_threshold=0.20)
rules = rules.sort_values(by='lift', ascending=False)
```

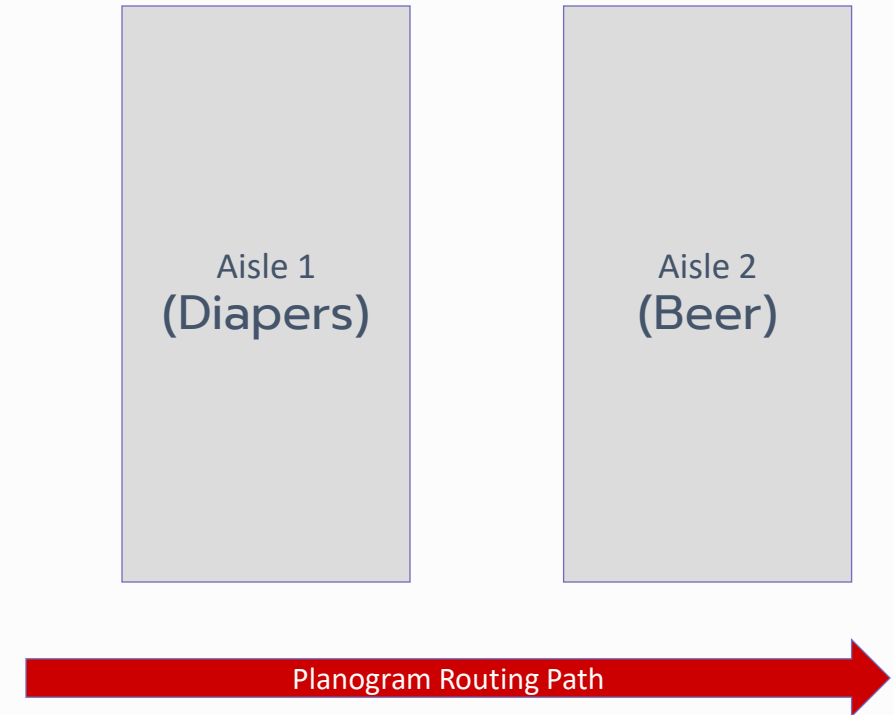
- Code explanation
 - Line 1: Import apriori and association_rules modules.
 - Line 4: Extract frequent itemsets with min_support threshold set at 1.0%.
 - Line 7: Extract association rules using a confidence threshold of 20%.
 - Line 8: Sort rules by lift to prioritize strong associations.

Module 7: Strategic Decision Framework

Translating model rules into physical layout changes and digital promotional levers

Physical Layouts

- Physical store layouts
 - Represent irreversible capital investments and store layout changes.
 - High physical setup cost requires high-confidence rules to justify execution.
- Shelf alignment
 - Placing diapers and beer in adjacent aisles alters the customer store route.
- Strategic risk
 - Incorrectly rearranging physical shelves generates an **asymmetric risk** of structural layout inefficiency.



Product Bundling

- Digital promotions
 - Represent reversible digital cross-selling recommendations at checkout.
 - Low operational cost enables rapid A/B testing of product combinations.
- Checkout carousel
 - Dynamically display complementary goods during the user check-out phase.
- Strategic risk
 - Over-recommending competing items yields an **asymmetric risk** of profit margin cannibalization.

Shopping Basket

- Burger (\$5.99)

Recommended Add-On:
 [+] French Fries (\$1.50)
 Save 20% when bought together

Strategic Tradeoffs

Decision matrix

Type 1 decisions require long observation windows and support $> 5\%$ to execute.

Type 2 decisions leverage low-support, high-lift rules for targeted email bundles.

Stop-loss trigger

Halt dynamic digital campaigns immediately if conversion probability falls below 2%.

Decision Tree

High Capital Risk?

```

|-- Yes => Type 1 Decision
|           (Planogram/Layout Change)
|-- No  => Type 2 Decision
|           (Digital Promo/Discount)
  
```

Fact-Based Management

- Analytical governance
 - Managers must establish clear KPIs before executing bundle recommendations.
 - Require monthly data audits to verify that lift factors remain stable.
- Operational rules
 - Establishing a **stop-loss trigger** to drop association rules if lift falls below 1.0.

Decision Tree

High Capital Risk?

```

|-- Yes => Type 1 Decision
|           (Planogram/Layout Change)
|-- No  => Type 2 Decision
|           (Digital Promo/Discount)
  
```

Module 8: Case Studies & Simulations

Hands-on application: manual cafe logs and large-scale retail transaction datasets

Case 1: Cafe Transaction Log

- Manual simulation
 - Students must manually build candidate frequent itemsets from a cafe ticket log.
 - Calculate support, confidence, and lift for 3 candidate basket combinations.
- Strategic exercise
 - Formulate a concrete shelf layout recommendation based on the calculated lift values.

TxID	Customer Basket Items
101	Coffee, Bread, Butter
102	Coffee, Butter
103	Tea, Bread, Jam
104	Coffee, Bread, Butter, Jam
105	Tea, Jam

Case 2: Large-Scale Retail Mining

- Programmatic mining
 - Students will ingest and analyze a dataset of online retail transactions.
 - Develop a Python script using mlxtend to run hyperparameter optimizations.
- Financial conversion
 - Translate top association rules into annual profit margin run rate increments.
- Data link
 - Dataset URL: <https://archive.ics.uci.edu/dataset/352/online+retail>